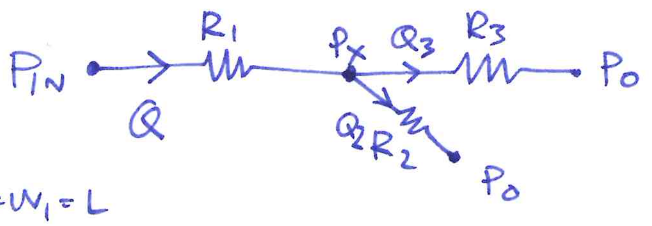
EX. 1

a)



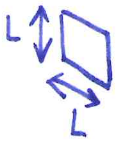
$V = 1\,\mu m/ms$

$H = W_1 = L$

$$Q = V \cdot H \cdot W_1 = \frac{1\,\mu m \cdot 30\,\mu m \cdot 30\,\mu m}{1\,ms} = 0.9 \cdot 10^{-12}\ m^3/s$$

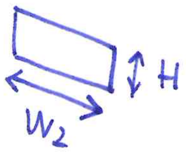
(54 ml/min)

VERY LOW FLOW RATE!



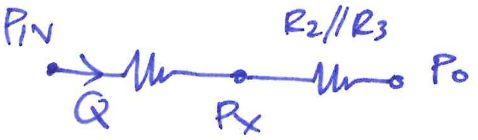
$$R_{eq} = \frac{2\ \text{Area}}{\text{perimeter}} = \frac{2L^2}{4L} = \frac{L}{2} = 15\,\mu m$$

$$R_1 = \frac{8\eta \cdot L_1}{\pi\, r_{eq}^4} = \frac{8 \cdot 10^{-3} \cdot 1\,cm}{\pi\,(15\,\mu m)^4} = 5 \cdot 10^{14}\ \left[\frac{Pa \cdot s}{m^3}\right]$$



$$\frac{W_2}{H} > 1.4 \Rightarrow R_3 = \frac{12\,\eta\,L_3}{\left(1 - 0.63\,\frac{H}{W_2}\right) H^3 \cdot W_2} = 4.3 \cdot 10^{14}\ \left[\frac{Pa \cdot s}{m^3}\right]$$

$$R_2 = \frac{R_1}{2} = 2.5 \cdot 10^{14}\ \left[\frac{Pa \cdot s}{m^3}\right]$$



$$R_2 // R_3 = \frac{2.5 \cdot 4.3}{(2.5 + 4.3)} \cdot 10^{14} = 1.6 \cdot 10^{14}$$

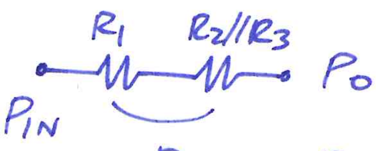
$$P_x = P_0 + R_2 // R_3 \cdot Q = 101\,kPa + \underbrace{0.9 \cdot 10^{-12}\,\frac{m^3}{s} \cdot 1.6 \cdot 10^{14}\,\frac{Pa \cdot s}{m^3}}_{144\ Pa}$$

$$= 101.14\ kPa$$

$$P_{IN} = P_x + Q \cdot R_1 = 101.14\ kPa + \underbrace{0.9 \cdot 10^{-12}\,\frac{m^3}{s} \cdot 5 \cdot 10^{14}}_{450\ Pa} =$$

$$= \boxed{101.59\ kPa}$$

b) $\eta_{BLOOD} = 4 \cdot \eta_{WATER}$ $\longrightarrow$



$R_{TOT} = R_1 + R_2 // R_3$

$$R\big|_{BLOOD} = 4 \cdot R_{TOT} = 4\,(R_1 + R_2 // R_3)$$

$$P_{IN} = P_0 + \underbrace{Q \cdot 4\,(R_1 + R_2 // R_3)}_{2.37\ kPa} = \boxed{103.37\ kPa}$$

(No pressure problems)

c) $Q_3 = \dfrac{P_x - P_o}{R_3} = \dfrac{144\ Pa}{4.3\cdot 10^{14}\left[\frac{Pa\cdot s}{m^3}\right]} \approx 0.33\cdot 10^{-12}\ m^3/s$

$Q_2 = \dfrac{P_x - P_o}{R_2} \approx \dfrac{144\ Pa}{2.5\cdot 10^{14}} = 0.57\cdot 10^{-12}\ m^3/s$

$(Q_2 + Q_3 = Q = 0.9\cdot 10^{-12}$ OK$)$

Approx: $t_1 = \dfrac{L_1\cdot W_1\cdot H}{Q} \approx \dfrac{1cm\cdot 30\mu m\cdot 30\mu m}{Q} = 10\ sec$

tot:

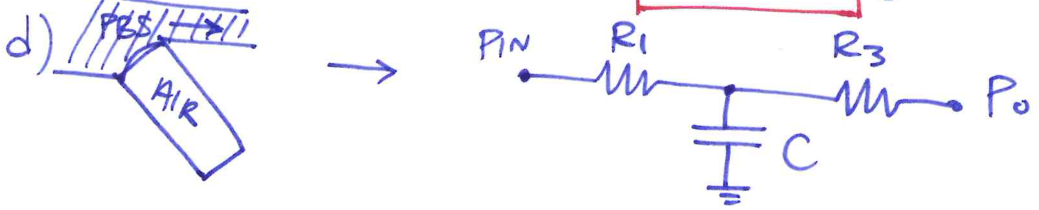
$t_1 + \max(t_2, t_3)$

$t_2 = \dfrac{0.5cm\cdot 30\mu m\cdot 30\mu m}{Q_2} = 7.9\ sec$

$t_3 = \dfrac{4cm\cdot 30\mu m\cdot 60\mu m}{Q_3} = 218\ sec$

$t_{TOT} = 10\ sec + 218\ sec = \boxed{228\ sec}$ (3.8 minutes)

d)



$\beta = \dfrac{1}{Vol}\cdot\dfrac{\Delta Vol}{\Delta P} \Rightarrow C = \dfrac{\Delta Vol}{\Delta P} = \beta\cdot Vol$

$Vol(L_2) = L_2\cdot W_1\cdot H = 0.5cm\cdot 30\mu m\cdot 30\mu m = 4.5\cdot 10^{-12}\ m^3$

$C = 7\cdot 10^{-6}\ \dfrac{1}{Pa}\cdot 4.5\cdot 10^{-12}\ m^3 = 31.5\cdot 10^{-18}\ \dfrac{m^3}{Pa}$

$\tau = (R_1 /\!/ R_3)\cdot C = \underbrace{\dfrac{5\cdot 4.3}{(5+4.3)}}_{2.3}\cdot 10^{14}\ \dfrac{\cancel{Pa}\cdot s}{\cancel{m^3}}\cdot 31.5\cdot 10^{-18}\ \dfrac{\cancel{m^3}}{\cancel{Pa}} =$

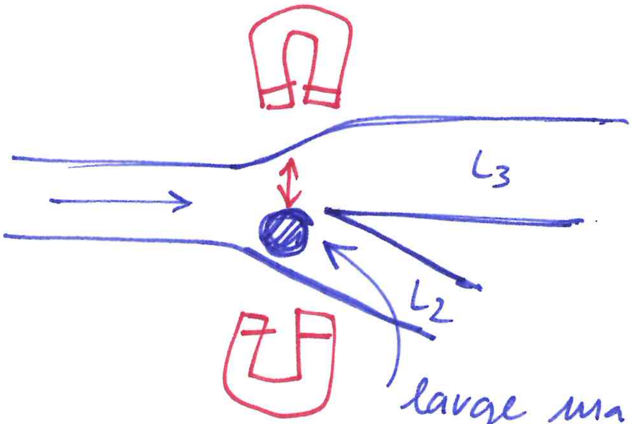
$= 2.3\cdot 10^{14}\cdot 31.5\cdot 10^{-18} = \boxed{7.2\ ms}$

e) $\eta_{\%} = \frac{Q_3}{Q} = \frac{0.33 \cdot 10^{-12}}{0.9 \cdot 10^{-12}} = \underline{36.6\ \%}$

low efficiency
↓
need for the actuator

SOLUTION:
(increase L_2 to increase R_2 and thus reduce Q_2)

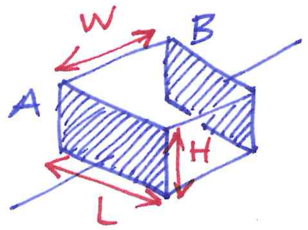
f)



large magnetic bead normally closing L_2 driven by external electromagnet.

EX. 2

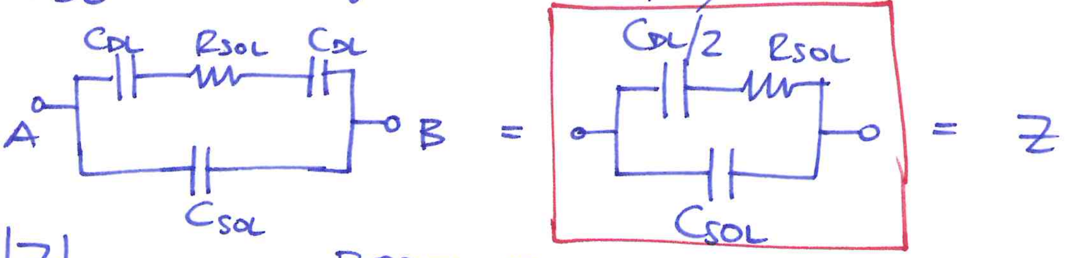
a)



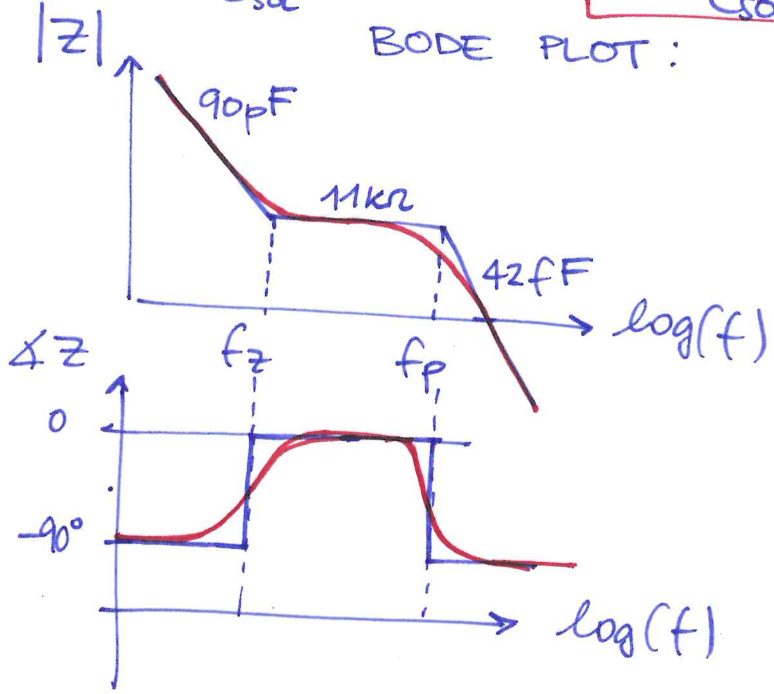
$$R_{SOL} = \rho \cdot \frac{W}{LH} = 0.66\,\Omega m \; \frac{30\mu m}{30\mu m \cdot 60\mu m} = 11\,k\Omega$$

$$C_{SOL} = \varepsilon_0 \cdot \varepsilon_{PBS} \cdot \frac{LH}{W} = \varepsilon_0\, 80 \cdot \frac{30\mu m \cdot 60\mu m}{30\mu m} = 42.5\,f$$

$$C_{DL} = \text{Area} \cdot C_0 = L \cdot H \cdot 0.1\,pF/\mu m^2 = 60 \cdot 30 \cdot 0.1\,pF = 180\,pF$$



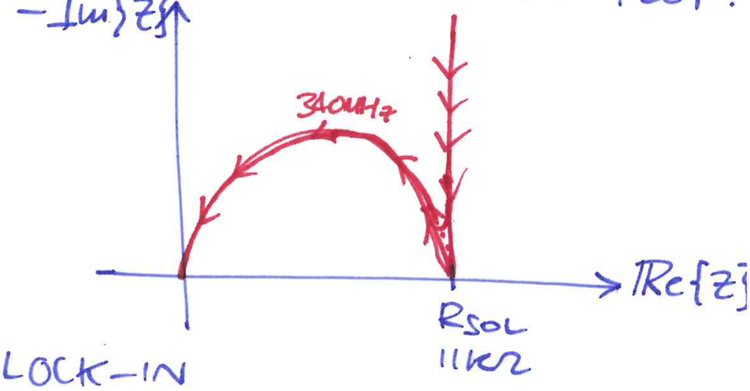
BODE PLOT:



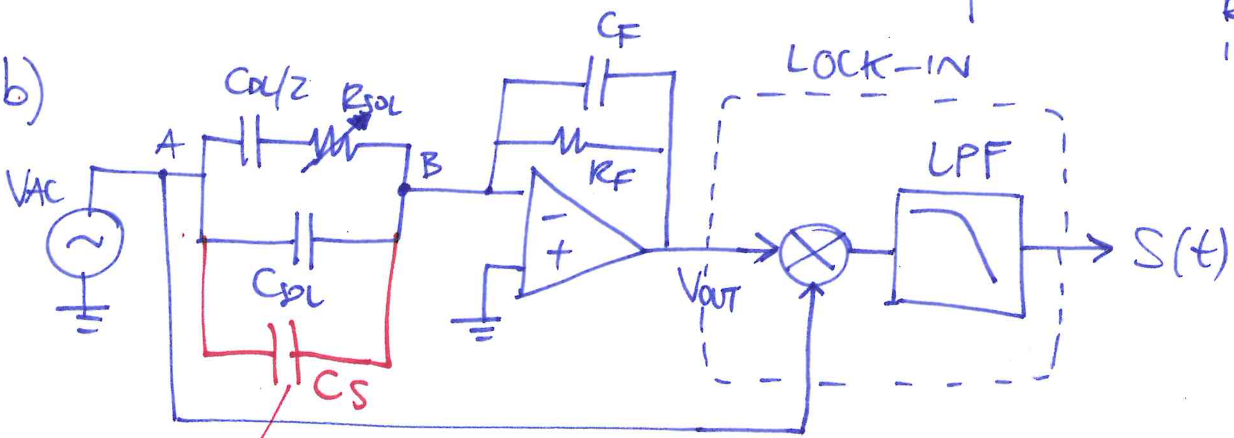
$$f_z = \frac{1}{2\pi R_{SOL} \cdot C_{DL}/2} = 160\,kHz$$

$$f_p = \frac{1}{2\pi R_{SOL} \cdot C_{SOL}} = 340\,MHz$$

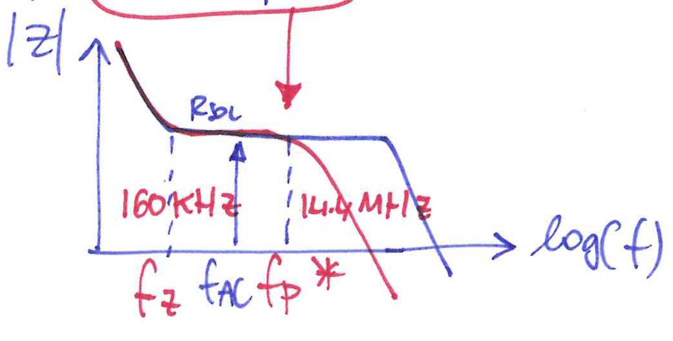
COLE-COLE PLOT:



b)



c) $C_S = 1\,pF$ → Reduces $f_p^* = \frac{1}{2\pi R_{SOL} \cdot C_S} = 14.4\,MHz$

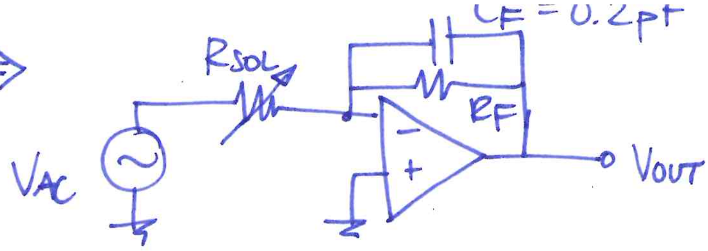


$$f_z < f_{AC} < f_p^*$$

$$160\,kHz < f_{AC} < 14.4\,MHz$$

$$f_{AC} = 10 \cdot f_z = 1.6\,MHz.$$

d) $f_{AC} \Rightarrow$



$$V_{OUT} = -\frac{V_{AC}}{R_{SOL}} \cdot R_F \Rightarrow \quad V_{OUT}\big|_{max} = 5V = \frac{0.1V}{11k\Omega} \cdot R_F$$

$$R_F = \frac{5V}{0.1V} \cdot 11k\Omega = \boxed{550k\Omega}$$ full swing (no margin that might be taken...)

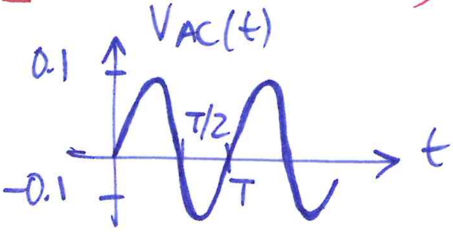
↓ CHECK :

$$f_{BW}\big|_{opamp} \sim \frac{1}{2\pi R_F C_F} = \frac{1}{2\pi \cdot 550k\Omega \cdot 0.2pF} \approx 1.4\,MHz \sim f_{AC}$$

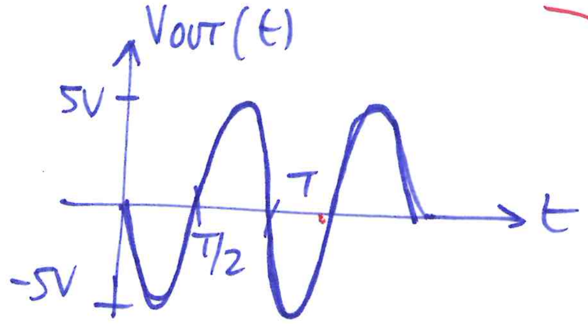
NO GOOD!

the pole @ f_{AC}

(another reason to reduce R_F in a real case)



$$T = \frac{1}{1.6MHz} = 625\,ns$$

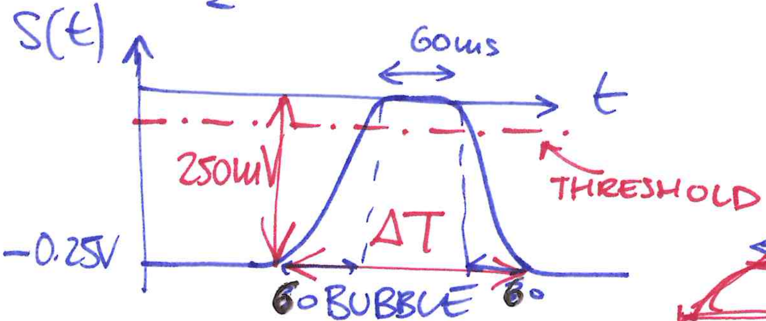


e)

$$S(t) = \underbrace{-\frac{V_{AC}}{R_{SOL}} \cdot R_F}_{V_{OUT}} \cdot V_{AC} \cdot \sin(2\pi f_{AC} t) \cdot \sin(2\pi f_{AC} t)$$

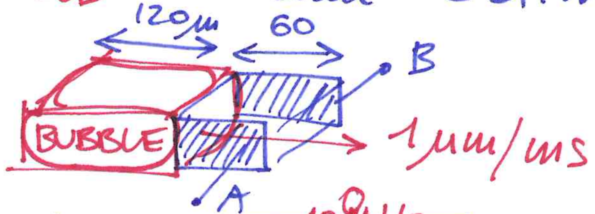
$$= -V_{AC}^2 \cdot \frac{R_F}{R_{SOL}} \cdot \frac{1}{2}\Big[\underbrace{\cos(0)}_{1} - \cancel{\cos(2\pi \cdot 2 f_{AC} t)}\Big] \quad \text{LPF}$$

$$= -\frac{1}{2} \cdot (0.1V)^2 \cdot 50 = -0.25V$$



S = 0 when air fills all the channel

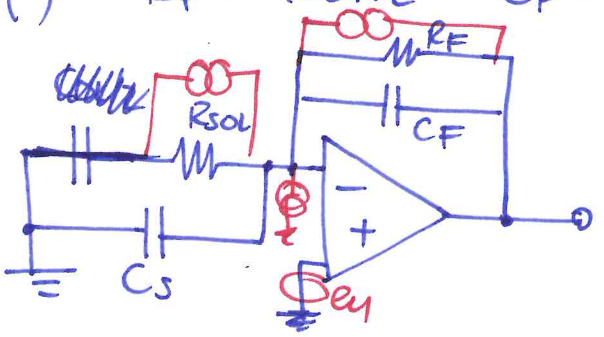
$V_{BUBBLE} = 2 \cdot LHW$



$$\Delta T = \frac{\cancel{120}\ 180\,\mu}{1\mu/ms} = 180\,ms \rightarrow \boxed{BW_{LPF} \sim \cancel{1kHz}\ 100Hz}$$

$$\tau = \frac{1}{2\pi \cdot 100Hz} = \cancel{0.16ms}\ 1.6ms \text{ or}$$

f) $R_F = 100\,k\Omega$ $\quad C_F = 0.5\,pF$ $\quad C_S = 1\,pF$



Noise terms:

- $\sqrt{\dfrac{4kT}{R_F}} = 0.4\ pA/\sqrt{Hz}$
- $\sqrt{\dfrac{4kT}{R_{SOL}}} = 1.2\ pA/\sqrt{Hz}$
- $2\pi f \underset{\substack{\parallel \\ (C_F+C_S)}}{C_{TOT}} \cdot e_n$
- $\dfrac{e_n}{R_F} = \dfrac{5\,nV/\sqrt{Hz}}{100\,k\Omega}$ negligible
- $\dfrac{e_n}{R_{SOL}} = \dfrac{5\,nV/\sqrt{Hz}}{11\,k\Omega} = 0.45\ pA/\sqrt{Hz}$

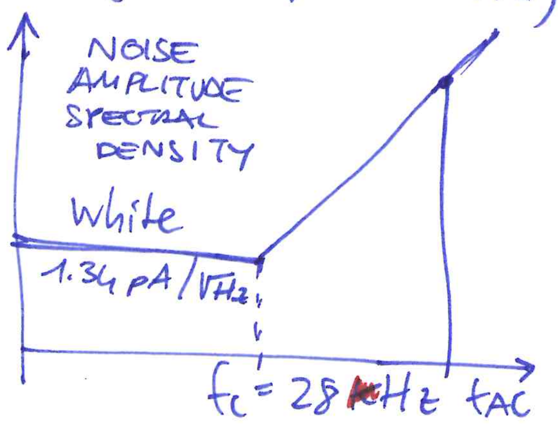
TOT WHITE TERMS:

$$\sqrt{0.4^2 + 1.2^2 + 0.45^2} = 1.34\ pA/\sqrt{Hz}$$

CORNER:

$$2\pi f_c \cdot 1.5\,pF \cdot 5\,nV/\sqrt{Hz} = 1.34\ pA/\sqrt{Hz}$$

$$f_c = \frac{1.34\ pA/\sqrt{Hz}}{(2\pi \cdot 1.5\,pF \cdot 5\,nV/\sqrt{Hz})} = 28.4\ \cancel{k}Hz$$



@ f_{AC}, $\underline{e_n \text{ dominates}}$

(f_{AC} should be reduced!!)

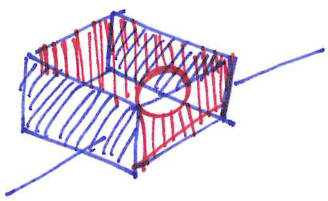
look for: max SNR(f)...

g) $\sigma_{NOISE} = 20\,ppm$ $\qquad \Delta Vol\big|_{min} \cong Vol \cdot 20\,ppm$

$= W \cdot H \cdot L \cdot 20\,ppm =$

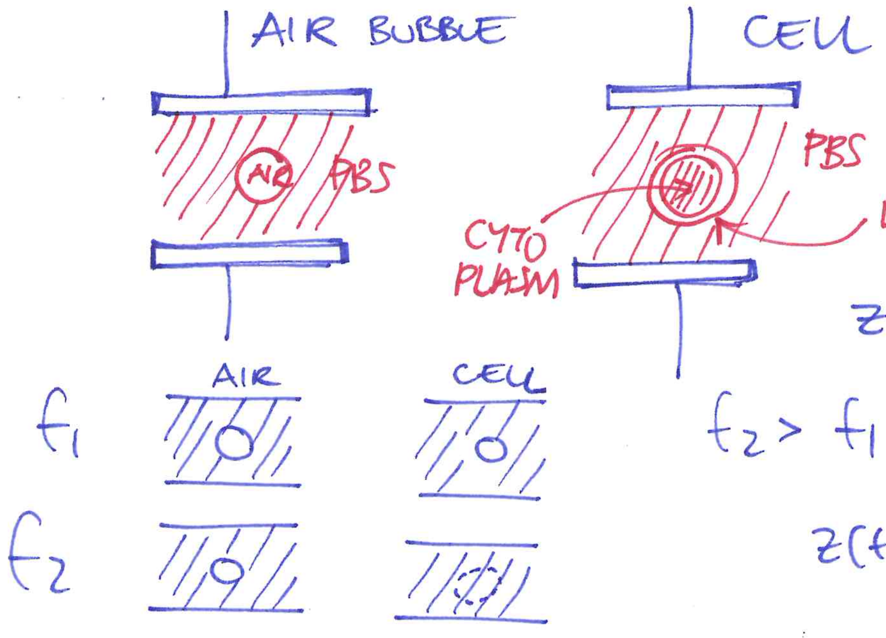
$= 60 \cdot 30 \cdot 30 \cdot (\mu m)^3 \cdot 20\,ppm$

$= 1\ \mu m^3$



$\frac{4}{3}\pi r^3 = 1\,\mu m^3 \Rightarrow$ $\boxed{r = 0.62\,\mu m}$ MIN. DIAMETER

h) YES, it is possible: USING more frequencies to shunt the membrane capacitance.



$f_2 > f_1$